

ESTEVAN, SK, Canada

WMO: 718620

Lat: 49.2050N

Lon: 102.9670W

Elev: 1904

StdP: 13.71

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-20.8	-34.0	0.9	-25.6	-29.0	1.2	-20.6	32.2	12.0	29.4	8.8	9.7	300	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	88.0	67.6	84.7	66.1	81.5	64.7	70.8	82.5	68.7	80.4	66.7	78.1	11.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	106.6	77.4	64.6	97.7	74.8	62.6	91.0	72.7	36.0	82.4	34.1	80.5	32.4	78.0	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
27.9	24.2	21.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-32.9	94.2	6.2	3.7	-37.4	96.9	-41.0	99.0	-44.5	101.1	-49.0	103.8
			-33.1	75.1	6.1	2.3	-37.5	76.8	-41.1	78.1	-44.5	79.4	-49.0	81.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	7.1	10.3	23.4	40.0	52.1	61.5	66.9	65.4	55.7	41.2	25.1	11.9	(6)
(7)		DBStd	23.95	15.43	14.54	14.30	10.53	8.25	6.39	5.59	6.23	8.29	9.40	12.27	14.09	(7)
(8)		HDD50	5926	1329	1112	824	323	74	3	0	36	298	746	1181		(8)
(9)		HDD65	9901	1794	1532	1289	750	407	138	45	72	295	737	1196	1646	(9)
(10)		CDD50	1745	0	0	1	23	138	349	524	477	207	26	0	0	(10)
(11)		CDD65	245	0	0	0	0	6	35	104	84	16	0	0	0	(11)
(12)		CDH74	3305	0	0	0	16	157	441	1142	1169	362	18	0	0	(12)
(13)		CDH80	997	0	0	0	1	34	106	340	406	108	2	0	0	(13)
(14)	Wind	WSAvg	10.5	11.3	11.0	11.3	11.6	11.6	10.2	8.6	8.6	9.3	10.4	10.7	10.9	(14)
(15)	Precipitation	PrecAvg	17.1	0.8	0.6	0.8	1.3	2.2	2.8	2.4	2.2	1.7	0.9	0.6	0.7	(15)
(16)		PrecMax	27.8	2.2	1.7	1.9	4.3	6.7	6.4	6.7	5.9	4.6	3.2	1.2	1.5	(16)
(17)		PrecMin	9.3	0.1	0.1	0.0	0.3	0.1	0.4	0.4	0.1	0.3	0.0	0.0	0.1	(17)
(18)		PrecStd	4.4	0.5	0.4	0.6	1.0	1.5	1.5	1.8	1.5	1.1	0.8	0.3	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	42.9	60.8	76.3	84.8	88.2	91.1	93.5	88.3	76.7	59.5	43.2	(19)
(20)		MCWB	35.7	37.7	46.8	55.2	60.3	67.9	70.5	68.2	65.1	56.4	47.2	36.8	(20)	
(21)		2%	DB	36.1	37.2	52.6	68.3	78.4	83.0	87.1	88.5	83.0	68.1	51.4	38.2	(21)
(22)		MCWB	33.0	33.9	43.2	51.3	58.1	65.3	69.3	66.9	62.2	53.0	42.6	34.1	(22)	
(23)		5%	DB	33.1	33.7	45.6	63.4	73.5	78.9	83.8	84.5	78.1	61.9	45.6	34.5	(23)
(24)		MCWB	30.9	31.4	39.3	48.5	55.3	63.0	68.0	65.9	60.2	49.6	39.1	31.7	(24)	
(25)		10%	DB	28.7	30.6	39.9	58.0	68.9	75.4	80.5	80.8	72.3	56.2	40.5	30.8	(25)
(26)		MCWB	27.0	28.9	36.0	45.9	53.3	61.1	66.3	64.5	57.4	46.9	35.8	28.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	38.0	48.8	55.9	63.4	72.6	74.5	72.2	67.5	58.6	48.1	37.4	(27)
(28)		MDCB	39.6	42.0	60.1	72.6	78.1	84.4	84.6	85.0	81.4	71.8	58.3	42.6	(28)	
(29)		2%	WB	33.3	34.1	43.8	52.4	59.6	67.9	71.7	69.4	64.6	54.3	43.1	34.3	(29)
(30)		MDCB	35.9	36.8	52.2	66.2	73.1	77.8	82.8	83.0	78.4	66.0	50.1	37.8	(30)	
(31)		5%	WB	30.8	31.7	39.5	49.4	57.1	65.1	69.7	67.4	61.6	51.0	39.5	31.7	(31)
(32)		MDCB	33.1	33.6	45.0	61.4	69.5	75.3	80.5	80.5	74.8	59.8	45.2	34.4	(32)	
(33)		10%	WB	27.1	28.8	36.1	46.6	54.9	62.8	67.7	65.5	58.9	47.9	36.1	28.8	(33)
(34)		MDCB	28.8	30.6	39.7	56.6	66.0	72.4	78.5	77.7	70.1	55.6	40.1	30.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.2	18.4	18.1	21.9	23.5	20.7	23.0	25.4	25.1	21.3	17.7	17.0	(35)	
(36)		5% DB	MCDBR	19.8	18.5	24.6	32.1	32.1	27.6	27.7	31.6	34.4	32.2	25.5	20.4	(36)
(37)		MCWBR	17.2	15.7	16.3	17.4	15.5	13.4	13.3	14.1	16.5	18.5	17.4	16.9	(37)	
(38)		5% WB	MCDBR	19.4	18.0	23.1	30.2	26.9	23.4	24.4	27.5	29.9	29.4	24.2	20.0	(38)
(39)		MCWBR	17.1	15.4	15.8	17.1	14.0	12.7	13.1	13.6	15.3	17.4	16.8	16.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.256	0.253	0.273	0.326	0.341	0.356	0.364	0.377	0.332	0.288	0.256	0.240	(40)	
(41)		taud	2.348	2.398	2.497	2.404	2.412	2.409	2.408	2.350	2.445	2.547	2.500	2.422	(41)	
(42)		Ebn at Noon	259	291	303	293	292	286	282	272	276	275	262	253	(42)	
(43)		Edh at Noon	21	25	28	34	35	36	35	36	29	22	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	469	817	1231	1573	1852	1960	2120	1701	1272	769	455	336	(44)	
(45)		RadStd	45	78	106	102	131	134	108	120	115	82	62	32	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page